Bubble detachment from circular cavities and flat surfaces

Ianto Cannon[a†], Stefan Endres[b] Lutz Mädler[b] and Marc Avila[a]

Received 00th January 20xx, Accepted 00th January 20xx DOI: 10.1039/x0xx00000x

We determine the maximum stable volume of bubbles attached to flat surfaces as a function of the contact radius, contact angle, and capillary length. Using a shooting method to solve the Young–Laplace equation, we compute equilibrium bubble profiles and identify the conditions under which three detachment modes occur: axisymmetric necking from the cavity centre, lateral instability for large cavities, and spreading followed by axisymmetric necking from the surrounding substrate. The predicted detachment volumes agree well with experimental measurements of electrolytic, boiling, and carbonated bubbles, as well as pendant drops. Our results unify and extend previous analytical estimates and apply directly to pendant drops by symmetry.

1 Introduction

1.1 Motivation

Annually, around 10^{11} kg of hydrogen gas is used in industry, mostly to produce fertilisers via the Haber process and to reduce metal ores in steelmaking and aluminium production [1]. Currently, more than 95% of hydrogen is produced from fossil fuels, largely from natural gas via steam methane reforming [2], which releases significant quantities of CO_2. Diversifying away from fossil fuels is essential to maintain a stable Earth climate system and ensure the long-term survival of human society [3]. Water electrolysis driven by renewable electricity is the most promising alternative [1], splitting water into hydrogen and oxygen without direct CO_2 emissions. However, gas bubbles that form and remain attached to the electrode surface limit the efficiency of water electrolysis. These bubbles block the active area, impede ion transport, and increase ohmic resistance [4–7]. Understanding and controlling the conditions for bubble detachment is therefore essential to optimise electrode geometry and wettability and promote early departure.

1.2 Background

The scale over which surface tension σ balances fluid density contrast $\rho_{\text{out}} - \rho_{\text{in}}$ and gravity g is the capillary length [8],

$$\lambda \equiv \sqrt{\frac{\sigma}{(\rho_{\text{out}} - \rho_{\text{in}})g}}. \tag{1}$$

The capillary length sets the characteristic radius of curvature of the bubble interface due to gravity. Two distinct attachment modes are recognised in the literature [9]. Pinned bubbles have a fixed foot radius r_0 equal to the radius r_0 of the nucleation cavity; their foot angle ϕ_0 changes as the bubble inflates. Spreading bubbles have foot angle ϕ_0 fixed at the contact angle ϕ_0 of the system; their foot radius r_0 changes as the bubble inflates.

For pinned bubbles, the classical Tate volume [10] assumes that a bubble detaches when the foot angle reaches $\phi_0 = \pi/2$, balancing buoyancy and surface tension to give $V = 2\pi\lambda^2 r_0$. Chesters [11] refined this in the limit $r_0/\lambda \to 0$, also assuming detachment at $\phi_0 = \pi/2$. Subsequent experiments, however, show systematic deviations from both models [12–16], due to non-spherical interfaces and the onset of asymmetrical instabilities. Stability analyses indicate that the bubble becomes unstable to asymmetrical perturbations for cavity radii $r_0 > 3.219\lambda$ [17], and no stable bubble exists at fixed volume for $r_0 > 3.832\lambda$ [18]. For such large cavities, lateral detachment from the rim is observed experimentally [15].

The study of spreading bubble departure dates back to Fritz [19], who numerically estimated the maximum bubble volume as a function of contact angle ϕ_0, $V = 0.887\lambda^3\phi_0^3$. Phan et al. [20] proposed an empirical model for nucleate pool boiling. More recent models account for contact angle hysteresis [21], using advancing, receding [22,23], or dynamic [24] contact angles as inputs. Experiments, however, reveal systematic deviations from these predictions, particularly for bubbles on superhydrophobic or high-contact-angle surfaces [25].

1.3 Aims

Here we treat pinned and spreading bubbles together, rather than as separate problems. We address two fundamental questions:

Q1 How large does a bubble grow before it detaches?

Q2 How does it detach?

We consider a simplified system consisting of a circular cavity of radius r_0 containing an already-nucleated bubble, surrounded by a smooth horizontal surface with a single-valued, non-hysteretic contact angle $\phi_0$that is, the contact angle is independent of whether the bubble foot advances or recedes. The fluid is quiescent, so forces are purely hydrostatic, and the system is physically equivalent to a pendant drop. Throughout this paper we refer to it as a bubble, but our results apply equally to pendant drops. We determine the detachment size for pinned and spreading bubbles across the full range of cavity sizes, capillary lengths ($0 \leq r_0/\lambda \leq \infty$), and contact angles ($0 \leq \phi_0 \leq \pi$).

2 Methods

2.1 Governing equation

To calculate the bubble shape, we use the method developed by Fritz [19] to find the detachment volume of spreading bubbles with small ϕ_0; consider a bubble attached to a solid horizontal surface, as depicted in Figure 1; z denotes the height above the substrate. The interface

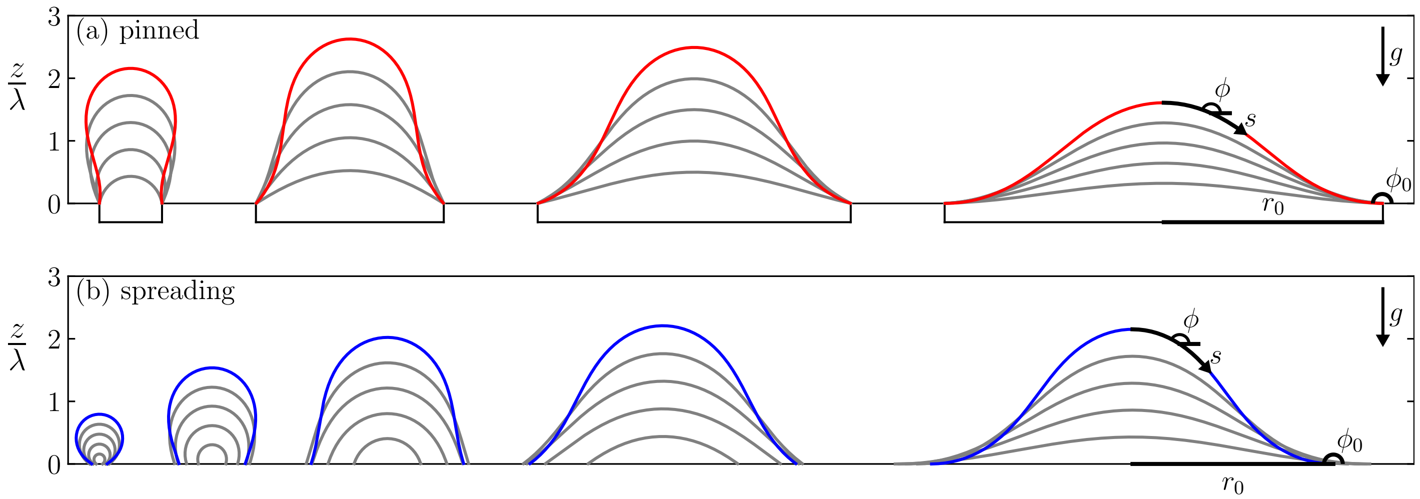

Figure 1 (a) Profiles of pinned bubbles with contact radius $r_0/\lambda \in \{0.5, 1.5, 2.5, 3.5\}$, from left to right. The largest pinned bubble for each contact radius is shown in red. (b) Profiles of spreading bubbles with contact angle $\phi_0/\pi \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$, from left to right. The largest spreading bubble for each contact angle is shown in blue. The coordinate system $(\phi, \phi_0, r_0, s, g)$ used for solving Equation 9 is shown in black.

of the bubble satisfies the Young–Laplace Equation[26]:

$$\sigma\left(\frac{1}{R_1}+\frac{1}{R_2}\right) = p_{\text{in}} - p_{\text{out}}, \tag{2}$$

where R_1 and R_2 are the principal radii of curvature at a point on the interface, and p_{in}, p_{out} are the pressures inside and outside the bubble, respectively.

Assuming both fluid phases are in hydrostatic equilibrium, the pressures at height z are given in terms of reference pressures $p_{0\text{in}}$ and $p_{0\text{out}}$ at the substrate $(z = 0)$:

$$p_{\text{in}}(z) = p_{0\text{in}} - \rho_{\text{in}} g z, \tag{3}$$
$$p_{\text{out}}(z) = p_{0\text{out}} - \rho_{\text{out}} g z. \tag{4}$$

Substituting into Equation 2 and dividing through by the surface tension gives:

$$\frac{1}{R_1}+\frac{1}{R_2} = \frac{p_{0\text{in}} - p_{0\text{out}}}{\sigma} + \frac{z}{\lambda^2}. \tag{5}$$

To eliminate the pressure difference $p_{0\text{in}} - p_{0\text{out}}$, we evaluate Equation 5 at the apex of the bubble, where $z = h$ and the interface is spherical, so $R_1 = R_2 = R_{\text{h}}$:

$$\frac{2}{R_{\text{h}}} = \frac{p_{0\text{in}} - p_{0\text{out}}}{\sigma} + \frac{h}{\lambda^2}. \tag{6}$$

Subtracting this from Equation 5 gives:

$$\frac{1}{R_1}+\frac{1}{R_2} = \frac{2}{R_{\text{h}}} + \frac{z-h}{\lambda^2}. \tag{7}$$

Assuming the bubble is axisymmetric about the vertical axis, we describe the interface shape in terms of arc length s from the apex, distance from the axis r, and the angle ϕ between the interface tangent and the horizontal. The two principal curvatures can be found through trigonometry[27]:

$$R_1 = \frac{ds}{d\phi}, \qquad R_2 = \frac{r}{\sin\phi}. \tag{8}$$

Substituting into Equation 7 and multiplying by the capillary length yields the governing equation for the bubble shape:

$$\lambda\frac{d\phi}{ds} + \frac{\lambda\sin\phi}{r} = \frac{2\lambda}{R_{\text{h}}} + \frac{z-h}{\lambda}. \tag{9}$$

This equation describes the curvature of an axisymmetric interface under gravity. All lengths appear scaled by the capillary length, implying self-similar bubble shapes and universal scaling with λ.

2.2 Numerical solution

We solve Equation 9 using a shooting method, initiating the integration at the apex $(r = 0,\ z = h,\ \phi = \pi)$ and sampling 10^4 logarithmically spaced initial guesses for R_h over $10^{-3}\lambda < R_h < 10^3\lambda$. The equations are integrated along the arc length s with a step size of $10^{-5}\lambda$ using the Adams–Bashforth method[27]. The complete set of calculations requires approximately 250 s on a single core of an M4 MacBook Pro. The code and accompanying data are freely available[28].

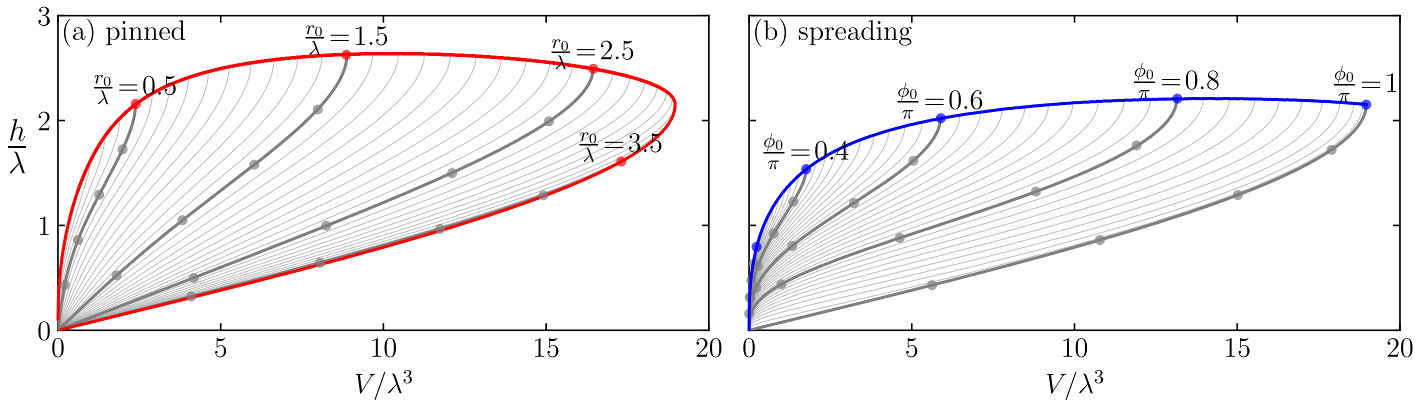

Figure 2 Bubble height h and volume V during evolution. (a) Grey lines: pinned bubbles with constant contact radius r_0; red line: locus of the maximum-volume pinned bubbles. (b) Grey lines: spreading bubbles with constant contact angle ϕ_0; red line: locus of the maximum-volume spreading bubbles. Grey and coloured circles correspond to the bubbles shown in Figure 1.

2.3 Boundary conditions for pinned and spreading bubbles

For pinned bubbles, the integration is terminated when $r = r_0$. For a given cavity radius r_0, this yields a family of equilibrium profiles parameterised by R_h, shown in Figure 1a. We assume that each profile nucleates as a flat interface spanning the cavity, and the foot angle decreases from $\phi_0 = \pi$ as the bubble inflates. For spreading bubbles, the integration is terminated when $\phi = \phi_0$. The resulting profiles are shown in Figure 1b, for a sequence of contact angles. Here we see the contact radii increase from $r_0 = 0$ as the bubbles inflate.

3 Results and Discussion

3.1 Pinned and spreading bubble evolution

To characterize the evolution of a growing pinned bubble, we plot the bubble height h as a function of volume V in Figure 2a, where the bubble volume

$$V = \int_0^h \pi r^2 \, dz, \tag{10}$$

is integrated using the midpoint rule. Each family of solutions in Figure 1a gives rise to a curve $h(V)$ in Figure 2a. Upon nucleation inside the cavity, the bubble has zero volume and zero height, corresponding to the bottom-left point in Figure 2a. As the bubble grows, its height h increases and it remains pinned on the $h(V)$ curve until one of three conditions is met:

1. If the foot angle ϕ_0 falls below the contact angle ϕ_0 of the system, the contact line unpins and spreads across the surface[9]. The bubble then follows an $h(V)$ curve in Figure 2b.

2. If the foot angle ϕ_0 reaches π, the interface becomes unstable to non-axisymmetric perturbations[17], the interface peak shifts to one side of the cavity, and the bubble detaches laterally. This occurs for $r_0 > 3.219\,\lambda$, at the right of Figure 2a.

3. If the bubble volume reaches a point where the $h(V)$ curve becomes vertical, the height can increase without a further increase in volume. The solution is no longer unique, and the interface becomes unstable. This leads to axisymmetric detachment. Each $h(V)$ curve in Figure 2a is therefore truncated at this point.

Figure 2b extends the analysis to spreading bubbles. It shows the possible volumes and heights for contact angles across the full range $0 \le \phi_0 \le \pi$. As with pinned bubbles, each curve reaches a point where $h(V)$ becomes vertical. This point defines the detachment volume of the spreading bubble, and the curves are truncated there. The detachment volume is seen in Figure 2b to increase with contact angle ϕ_0. The tallest possible bubble has $h = 2.642\,\lambda$, which is considerably less than the analytical limit of $h = 3.42\,\lambda$ established for two-dimensional columnar pendant drops[29].

3.2 Detachment volume of pinned bubbles

Figure 3(a) shows the detachment volume V_p of pinned bubbles. The detachment volume increases with cavity radius, following a trend close to the Tate prediction[10]; deviations arise due to the non-spherical interface shape under buoyancy. These deviations are confirmed by experimental measurements of bubbles emitted from tubes in water[13], bubbles in carbonated water from millimetric cavities[14], and pendant drops[16]. The solid line in Figure 3c shows the foot angle ϕ_0 at departure. Contrary to Chesters' assumption[11], the foot angle at detachment exceeds $\pi/2$ for all cavity radii. For cavities with $r_0 > 3.219\,\lambda$, ϕ_0 reaches π and the interface becomes unstable to non-axisymmetric perturbations[17], causing the bubble to slide to one side and detach laterally. No stable solution exists for $r_0 > 3.832\,\lambda$[18]; above this radius, Equation 9 has no solution with $\phi_0 < \pi$ and lateral detachment is immediate, consistent with the dripping behaviour reported in sessile drop experiments[15]. Figure 3c also shows the minimum foot angle reached during bubble evolution. If the surface contact angle ϕ_0 exceeds this minimum, the bubble will become unpinned and spread across the surface. The minimum angle closely follows the fit $\phi_0/\pi = \sqrt{r_0/3.5\lambda}$.

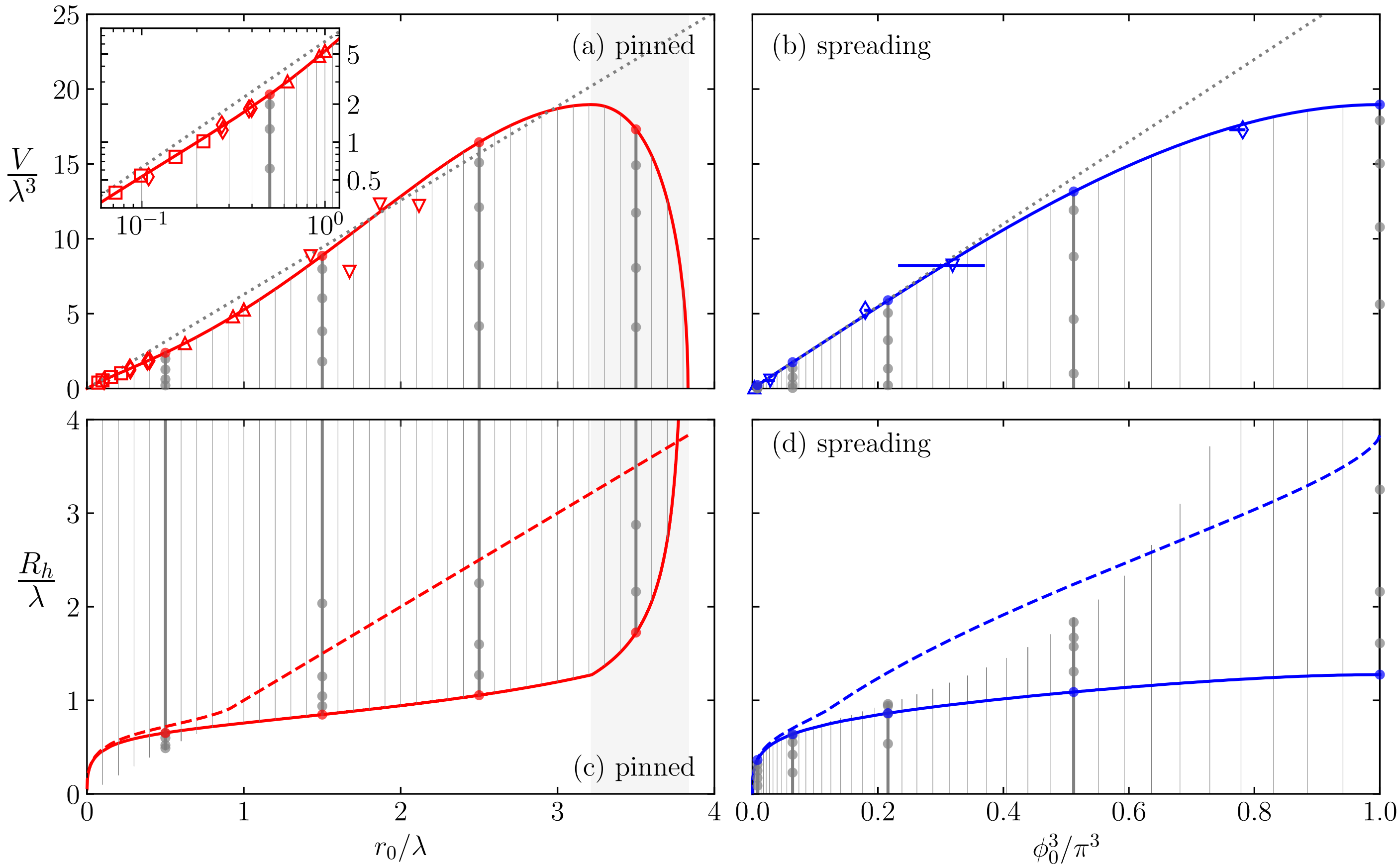

Figure 3 (a) Red line: detachment volume V of pinned bubbles as a function of cavity radius r_0. The dotted grey line, $V = 2\pi\lambda^2 r_0$, is the Tate prediction[10]. Empty symbols denote experimental data: squares[13], diamonds[14], down-triangles[15], and up-triangles[16]. Inset: same data on a logarithmic scale, showing the small-radius range in greater detail. (b) Detachment volume V of spreading bubbles as a function of contact angle ϕ_0, cubed to allow comparison with the Fritz[19] prediction $V = 0.887\lambda^3\phi_0^3$ (dotted grey line). Symbols denote experimental data: up-triangles[23], down-triangles[22], and diamonds[25]; error bars indicate advancing and receding contact angles. (c) Solid red line: apex radius of curvature R_h of pinned bubbles at detachment; dashed red line: the maximum horizontal radius $r_{\max}$ reached by pinned bubbles. (d) Solid blue line: apex radius of curvature R_h of spreading bubbles at detachment; dashed red line: the maximum horizontal radius $r_{\max}$ reached by spreading bubbles. Grey solid lines and filled circles show the evolution of bubbles up to detachment.

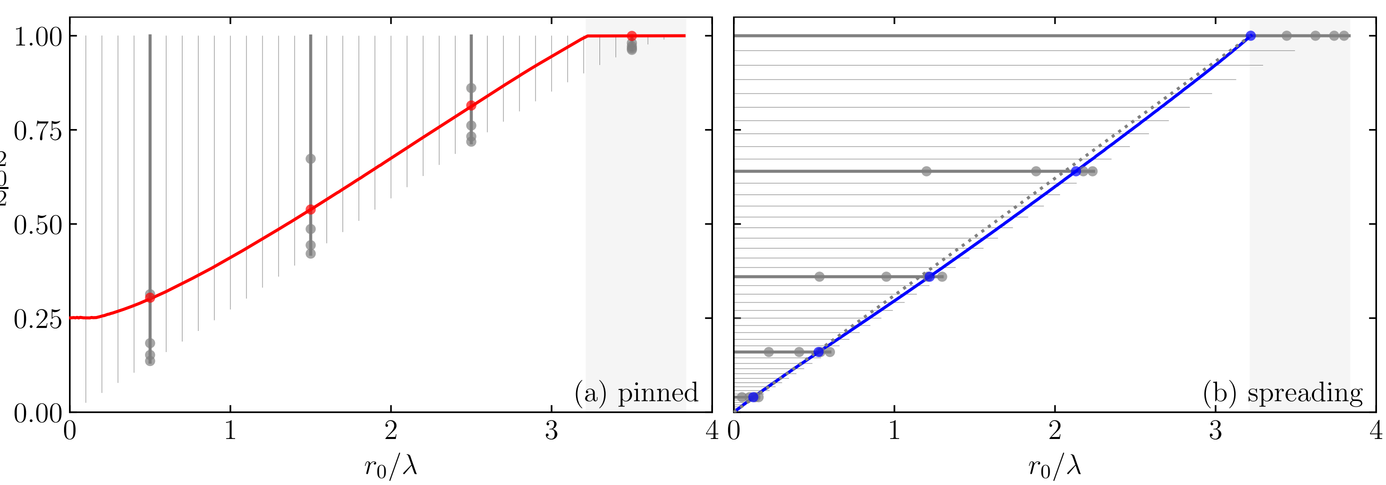

Figure 4 (a) Foot angle ϕ_0 at detachment (red dots) and minimum foot angle during evolution (red line). The dashed black line shows the fit $\phi_0/\pi = \sqrt{r_0/3.5\lambda}$. The shaded region $r_0 > 3.219\lambda$ indicates where ϕ_0 reaches π and the bubble becomes unstable to asymmetric perturbations. (d) Maximum foot radius r_0 during bubble evolution (blue line) and foot radius at detachment (blue dots). The dashed black line shows the fit $r_0 = 3.219\lambda(\phi_0/\pi)^2$.

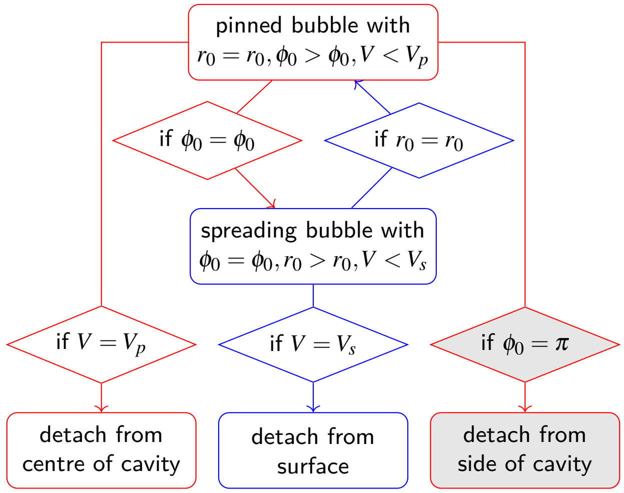

Figure 5 Decision tree for bubble detachment. Red branches correspond to pinned bubbles and blue branches to spreading bubbles. The quantities V_p and V_s denote the corresponding maximum stable bubble volumes, while ϕ_0 is the contact angle on the surrounding substrate.

Table 1 Key properties of significant bubbles, expressed in units of the capillary length λ. Shown are the foot angle ϕ_0, foot radius r_0, apex height h, apex radius of curvature R_h, volume V. Extremal quantities are highlighted in bold

Significance	ϕ_0/π	r_0/λ	h/λ	R_h/λ	V/λ^3
Largest pinned bubble	1.000	3.219	2.152	1.274	**18.96**
Largest spreading bubble	1.000	3.219	2.152	1.274	**18.96**
Tallest pinned bubble	0.7750	1.720	**2.642**	0.8867	10.62
Tallest spreading bubble	0.8330	2.297	**2.213**	1.126	14.48
Largest stable foot radius	1.000	**3.832**	0.000	∞	0.000

3.3 Detachment volume of spreading bubbles

Figure 3b shows the detachment volume V_s of spreading bubbles. For small contact angles, the Fritz prediction is accurate. As $\phi_0 \to \pi$, however, the detachment volume saturates below Fritz's extrapolation at a maximum of $V = 18.96\lambda^3$. Our results are confirmed by measurements of water electrolysis[23], boiling[22], and air bubbles over superhydrophobic surfaces[25]. Figure 3d shows the radius of the bubble foot at departure, which closely follows the fit $r_0 = 3.219\lambda(\phi_0/\pi)^2$. Figure 3d also shows the maximum radius of the foot during bubble evolution. In all cases, the maximum foot radius exceeds the departure radius, indicating that the contact line initially spreads and then partially recedes prior to detachment. In some instances[12], the foot radius may again equal the cavity radius r_0, re-pinning the bubble at the cavity, in which case the departure volume corresponds to V_p.

3.4 Pinning and unpinning of the contact line

Figure 5 gives the decision tree for bubble evolution up to detachment. Given ϕ_0, r_0, and the capillary length λ of a specific system, the detachment size is determined by following this tree, using the functions plotted in Figure 3. Table 1 gives the properties of the limiting bubble configurations, where we see that the largest pinned and spreading bubbles in fact correspond to the same solution of Equation 9. Note that we measure the bubble volume above the contact line; some gas may remain attached to the surface after detachment, so the true released volume may be slightly less.

4 Conclusions

4.1 Answers to the two key questions

We now return to the two questions posed at the outset.

Answer to Q1. The detachment volume depends on whether the bubble is pinned or spreading. For pinned bubbles, the detachment volume rises nearly linearly with cavity radius, closely following the Tate prediction $V = 2\pi\lambda^2 r_0$, with quantitative corrections arising from the non-spherical interface shape under buoyancy (Figure 3). For spreading bubbles, the detachment volume increases monotonically with contact angle and saturates at $V_s = 18.96\lambda^3$ as $\phi_0 \to \pi$, below the Fritz extrapolation (Figure 3b). Both results are confirmed by independent experimental datasets spanning electrolytic, boiling, and carbonated bubbles, as well as pendant drops.

Answer to Q2. The detachment mode is determined by which threshold is reached first during bubble growth (Figure 5). A pinned bubble detaches axisymmetrically from the cavity centre when its volume reaches the maximum of the r_0 curve; it detaches laterally when the foot angle reaches $\phi_0 = \pi$, which occurs only for $r_0 > 3.219\lambda$; and it transitions to spreading mode when the foot angle ϕ_0 falls below the contact angle ϕ_0 on the surrounding substrate. A spreading bubble then detaches from the surface when its volume reaches V_s. No stable bubble exists for $r_0 > 3.832\lambda$, above which lateral detachment is immediate.

4.2 Outlook

In summary, we have computed the maximum stable volume of axisymmetric bubbles on flat surfaces as a function of contact radius and contact angle, using a numerical solution of the Young–Laplace equation. The results unify the pinned and spreading detachment modes within a single framework, recover the classical Tate and Fritz limits, and identify the thresholds at which non-axisymmetric instabilities govern departure. All results scale universally with the capillary length λ and apply equally to pendant drops. The numerical code used in this study is freely available[28], and can provide an exact detachment criterion for full electrolyser simulations such as that of Ross *et al.*[7], as well as quantitative guidance for designing electrodes with optimised wettability and cavity geometry to promote early bubble detachment and improve electrolysis efficiency. In future work, the model can be extended to account for heterogeneous surfaces, which give rise to contact angle hysteresis.

5 Data availability

The data and code that support the findings of this article are openly available on Zenodo at `https://doi.org/10.5281/zenodo.20720640` (ref. 28).

6 Author Contributions

I. Cannon: conceptualisation, software, formal analysis, visualisation, writing – original draft. S. Endres: software, writing – review & editing. M. Avila: conceptualisation, methodology, writing – review & editing. L. Mädler: conceptualisation, methodology, writing – review & editing.

7 Conflicts of interest

There are no conflicts to declare.

8 Acknowledgements

This research was supported by the University of Bremen and State of Bremen within the Humans on Mars Initiative and partially supported by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy EXC-3036 The Martian Mindset, project number: 533607631. I. Cannon gratefully acknowledges financial support from the Alexander von Humboldt Foundation through a Humboldt Research Fellowship for Postdoctoral Researchers.

Notes and references

[1] F. Kullmann, J. Linßen and D. Stolten, The Role of Hydrogen for the Defossilization of the German Chemical Industry, *Int. J. Hydrogen Energy*, 2023, **48**, 38936–38952, DOI: 10.1016/j.ijhydene.2023.04.191.

[2] G. Franchi, M. Capocelli, M. D. Falco, V. Piemonte and D. Barba, Hydrogen Production via Steam Reforming: A Critical Analysis of MR and RMM Technologies, *Membranes*, 2020, **10**, 10, DOI: 10.3390/membranes10010010.

[3] W. Steffen, J. Rockström, K. Richardson, T. M. Lenton, C. Folke, D. Liverman, C. P. Summerhayes, A. D. Barnosky, S. E. Cornell, M. Crucifix, J. F. Donges, I. Fetzer, S. J. Lade, M. Scheffer, R. Winkelmann and H. J. Schellnhuber, Trajectories of the Earth System in the Anthropocene, *Proc. Natl. Acad. Sci.*, 2018, **115**, 8252–8259, DOI: 10.1073/pnas.1810141115.

[4] H. Vogt, The Problem of the Departure Diameter of Bubbles at Gas-Evolving Electrodes, *Electrochim. Acta*, 1989, **34**, 1429–1432, DOI: 10.1016/0013-4686(89)87183-X.

[5] A. Angulo, P. van der Linde, H. Gardeniers, M. Modestino and D. F. Rivas, Influence of Bubbles on the Energy Conversion Efficiency of Electrochemical Reactors, *Joule*, 2020, **4**, 555–579, DOI: 10.1016/j.joule.2020.01.005.

[6] B. Ross, S. Haussener and K. Brinkert, Impact of Gas Bubble Evolution Dynamics on Electrochemical Reaction Overpotentials in Water Electrolyser Systems, *J. Phys. Chem. C*, 2025, **129**, 4383–4397, DOI: 10.1021/acs.jpcc.5c00220.

[7] B. Ross, K. Skidmore, S. Haussener and K. Brinkert, Transient Simulation of Gas Bubble Evolution and Overpotential Dynamics for the Hydrogen Evolution Reaction, *ACS Electrochem.*, 2026, **2**, 113–123, DOI: 10.1021/acselectrochem.5c00291.

[8] P. S. Laplace, *Traité de Mécanique Céleste*, Courcier, Paris, 1798.

[9] A. K. Chesters, Modes of Bubble Growth in the Slow-Formation Regime of Nucleate Pool Boiling, *Int. J. Multiphase Flow*, 1978, **4**, 279–302, DOI: 10.1016/0301-9322(78)90003-4.

[10] T. Tate, On the Magnitude of a Drop of Liquid Formed under Different Circumstances, *Lond. Edinb. Dublin Philos. Mag. J. Sci.*, 1864, **27**, 176–180, DOI: 10.1080/14786446408643645.

[11] A. K. Chesters, An Analytical Solution for the Profile and Volume of a Small Drop or Bubble Symmetrical about a Vertical Axis, *J. Fluid Mech.*, 1977, **81**, 609–624, DOI: 10.1017/S0022112077002250.

[a] *Center of Applied Space Technology and Microgravity (ZARM), University of Bremen, 28359 Bremen, Germany.*
[b] *Leibniz Institute of Materials Engineering (IWT), University of Bremen, 28359 Bremen, Germany.*
[†] *E-mail: ianto.cannon@zarm.uni-bremen.de*

[12] D. Gerlach, G. Biswas, F. Durst and V. Kolobaric, Quasi-Static Bubble Formation on Submerged Orifices, *Int. J. Heat Mass Transfer*, 2005, **48**, 425–438, DOI: 10.1016/j.ijheatmasstransfer.2004.09.002.

[13] F. J. Lesage and F. Marois, Experimental and Numerical Analysis of Quasi-Static Bubble Size and Shape Characteristics at Detachment, *Int. J. Heat Mass Transfer*, 2013, **64**, 53–69, DOI: 10.1016/j.ijheatmasstransfer.2013.04.019.

[14] B. K. Mori and W. Baines, Bubble Departure from Cavities, *Int. J. Heat Mass Transfer*, 2001, **44**, 771–783, DOI: 10.1016/s0017-9310(00)00133-2.

[15] S. Sasetty and T. Ward, Stability and Critical Volume of a Suspended Pendant Drop in Air via Experiments and Eigenvalue Analysis, *Colloids Surf., A*, 2023, **666**, 131346, DOI: 10.1016/j.colsurfa.2023.131346.

[16] R. Gunde, A. Kumar, S. Lehnert-Batar, R. Mäder and E. J. Windhab, Measurement of the Surface and Interfacial Tension from Maximum Volume of a Pendant Drop, *J. Colloid Interface Sci.*, 2001, **244**, 113–122, DOI: 10.1006/jcis.2001.7916.

[17] D. H. Michael and P. G. Williams, The Equilibrium and Stability of Axisymmetric Pendent Drops, *Proc. R. Soc. Lond., Math. Phys. Sci.*, 1976, **351**, 117–127, DOI: 10.1098/rspa.1976.0132.

[18] J. A. F. Plateau, *Statique Expérimentale et Théorique Des Liquides Soumis Aux Seules Forces Moléculaires*, Paris, Gauthier-Villars, 1873.

[19] W. Fritz, Berechnung Des Maximalvolumes von Dampfblasen, *Phys., Zeitschr,*, 1935, **36**, 379–384.

[20] H. T. Phan, N. Caney, P. Marty, S. Colasson and J. Gavillet, Surface Wettability Control by Nanocoating: The Effects on Pool Boiling Heat Transfer and Nucleation Mechanism, *Int. J. Heat Mass Transfer*, 2009, **52**, 5459–5471, DOI: 10.1016/j.ijheatmasstransfer.2009.06.032.

[21] P. G. de Gennes, Wetting: Statics and Dynamics, *Rev. Mod. Phys.*, 1985, **57**, 827–863, DOI: 10.1103/RevModPhys.57.827.

[22] T. P. Allred, J. A. Weibel and S. V. Garimella, The Role of Dynamic Wetting Behavior during Bubble Growth and Departure from a Solid Surface, *Int. J. Heat Mass Transfer*, 2021, **172**, 121167, DOI: 10.1016/j.ijheatmasstransfer.2021.121167.

[23] Ç. Demirkır, J. A. Wood, D. Lohse and D. Krug, Life beyond Fritz: On the Detachment of Electrolytic Bubbles, *Langmuir*, 2024, **40**, 20474–20484, DOI: 10.1021/acs.langmuir.4c01963.

[24] A. Mukherjee and S. G. Kandlikar, Numerical Study of Single Bubbles with Dynamic Contact Angle during Nucleate Pool Boiling, *Int. J. Heat Mass Transfer*, 2007, **50**, 127–138, DOI: 10.1016/j.ijheatmasstransfer.2006.06.037.

[25] J. Huang and R. Li, Effects of Surface Wettability on Bubble Dynamics and Induced Liquid Flow: Finite-Difference Analysis of Two-Phase Particle Image Velocimetry, *Phys. Rev. Fluids*, 2026, **11**, 023603, DOI: 10.1103/jvxz-8mzv.

[26] P.-G. De Gennes, F. Brochard-Wyart and D. Quéré, *Capillarity and Wetting Phenomena*, Springer New York, New York, NY, 2004, DOI: 10.1007/978-0-387-21656-0.

[27] F. Bashforth and J. C. Adams, *An Attempt to Test the Theories of Capillary Action by Comparing the Theoretical and Measured Forms of Drops of Fluid*, Cambridge University Press, 1883.

[28] I. Cannon, *Bubble*, Zenodo, 2026, DOI: 10.5281/zenodo.20720640.

[29] P. T. Sumesh and R. Govindarajan, The Possible Equilibrium Shapes of Static Pendant Drops, *J. Chem. Phys.*, 2010, **133**, 144707, DOI: 10.1063/1.3494041.